Colby Threlkeld

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Education

Georgia State University — Expected May 2027
B.S. in Computer Science, GPA: 3.5 / 4.0 — *Atlanta, GA*

- **Relevant Coursework**: Artificial Intelligence, Machine Learning, Database Systems, Software Development, Probability & Statistics in R, Computing in Python, Data Structures & Algorithms, Robotics, Linear Algebra, Calculus I & II

Experience

Apple — Summer 2026
Software Engineer Intern – Apple Online Store Team (AOS) — *Austin, TX*

- Developed recommendation & warning feature for AOS using Apple Intelligence & user behavioral data (React, JavaScript)
- Cut LLM response latency by over 60% to meet production target & designed fallback model pipeline (Python, Interlinked)
- Resolved 4 major production bugs (JS, TS) affecting millions of users, owning each fix from code review through merging
- Quantified projected feature impact with data & product partners, pitching to leadership for an approved roadmap

Interdisciplinary Neuroscience for Social Development Lab (INSD) — Aug 2026 – Present
Research Assistant, Georgia State University — *Atlanta, GA*

- Built Python data pipelines that ingest, transform, & validate large-scale, high-dimensional screenshot data collected through the Stanford Screenomics research software for an IRB-approved study of adolescent smartphone use
- Supporting OCR & computer vision workflows (OpenCV, Tesseract) to extract text & app data from screenshots
- Collaborating with neuroscience & psychology researchers to integrate smartphone, MRI, & survey datasets

Progsu — Jan 2026 – Present
Director – Georgia State Programming Club — *Atlanta, GA*

- Built & shipped the Hacklanta website & application platform used to run a hackathon that drew 1,400+ applications & 700+ participants, handling registration, applicant review, team formation, & participant communications before & during the event
- Developed full-stack club wiki (wiki.progsu.com) serving 1,800+ members, with a GSU course explorer database & a career knowledge base of roadmaps covering DSA, system design, & Atlanta internship pipelines
- Secured & managed club partnerships with Anthropic, OpenAI, Google, Adobe, MLH, & more for sponsorships, speakers, & swag, & wrote educational tech content for social media outreach campaigns with the design team to grow membership

Broadwell Group Inc. (BGI) — Summer 2025
Frontend Engineer Intern — *Atlanta, GA*

- Replaced a 9-year-old legacy site with a new company website as the sole developer, designing & building it on custom WordPress with HTML, CSS, & JavaScript & taking it from first mockup to production launch
- Reduced annual hosting costs by at least 24% through auditing the legacy hosting package against current traffic & storage needs, selecting a modern provider, & migrating the domain, DNS, SSL certificate, & site content with no downtime
- Cut page load time by over 45% after rebuilding the frontend with a responsive mobile-first layout, compressing & lazy-loading images & scripts, & removing unused plugins that were slowing the legacy site on every page load

Projects

Global Fan Platform | *TypeScript, SQL, HTML, CSS, Next.js, Supabase, NextAuth, OpenAI API* — May 2026

- Built full-stack platform for live events (World Cup, Olympics) using Next.js, TypeScript, & PostgreSQL featuring event feeds, communities, messaging, trip planning, AI-powered event finder, & real-time voice-to-voice translator
- Engineered a speech-to-speech translation pipeline with 5 server-side API routes chaining OpenAI Whisper (STT), GPT translation, & TTS, including a streaming endpoint for low-latency cross-language conversation between fans at events

Icarus – Autonomous Drone | *Python, OpenCV, Raspberry Pi, MAVLink, Kalman Filter, PID* — Sep 2026 – Present

- Designed a sub-$250 autonomous quadcopter targeting $\leq$10 cm landing precision (vs. $\sim$4.9 m GPS error) by replacing a $260 RTK-GPS receiver with an $84 camera, optical-flow, & Pi Zero 2 W sensing stack, cutting positioning hardware cost $\sim$68%
- Specified input-validation & safety rules for every data stream: reject frames older than 0.2 s, unknown tag IDs, & pose jumps outside the filter gate; trigger hover/abort within 1 s on lost pad, vision below 10 fps, or low battery, with manual RC override
- Designed a Kalman filter on a Raspberry Pi Zero 2 W to fuse $\sim$30 fps camera pose of the landing pad with optical-flow velocity & ToF height, predicting the drone's position $\sim$0.1 s ahead so the vision delay doesn't become a 3+ cm landing error

Hybrid Fuzzy Logic Recommendation Engine Research Paper | *Python, PyTorch, Pandas, scikit-fuzzy* — Apr 2026

- Product recommender that re-ranks Item-CF candidates with fuzzy logic steered by user preferences (price, consumer trust, etc.)
- Fed 10 input variables (price sensitivity, brand loyalty) into a 74-rule Mamdani inference engine to score recommendations
- Designed system to produce explainable recommendations where each output can be traced to readable if-then rules
- Processed features from Amazon review data including sentiment scoring, price normalization, & user interaction history

Technical Skills

Languages: Python, JavaScript, TypeScript, C, Java, SQL, HTML, CSS, SCSS, R

Frontend: React.js, Next.js, Tailwind CSS, Figma, WordPress

Backend, Databases & APIs: Node.js, Express.js, MongoDB (Mongoose ODM), REST APIs, PostgreSQL, MySQL, Postman

Data, ML & DevTools: PyTorch, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Git/GitHub, Vercel, MCP